

Answer key | Lesson 5 - 3 x 3 digit multiplication

Mild

1) Copy and complete the long multiplication

- a) 213×124 : $852 + 4\,260 + 21\,300 = \mathbf{26\,412}$
b) 321×213 : $963 + 3\,210 + 64\,200 = \mathbf{68\,373}$

2) Complete the calculations

- a) $112 \times 213 = \mathbf{23\,856}$
b) $221 \times 132 = \mathbf{29\,172}$
c) $312 \times 211 = \mathbf{65\,832}$
d) $231 \times 121 = \mathbf{27\,951}$
e) $144 \times 212 = \mathbf{30\,528}$
f) $423 \times 121 = \mathbf{51\,183}$

Spicy

1) Copy and complete the long multiplication

- a) 341×214 : $1\,364 + 3\,410 + 68\,200 = \mathbf{72\,974}$
b) 523×136 : $3\,138 + 15\,690 + 52\,300 = \mathbf{71\,128}$

2) Complete using long multiplication

- a) $412 \times 325 = \mathbf{133\,900}$
b) $534 \times 241 = \mathbf{128\,694}$
c) $618 \times 152 = \mathbf{93\,936}$
d) $725 \times 364 = \mathbf{263\,900}$

3) Word problems

- a) $312 \times 24 = \mathbf{7\,488}$ dirhams
b) $213 \times 124 = \mathbf{26\,412}$ passengers
c) $143 \times 132 = \mathbf{18\,876}$ items

Hot

1) Complete using long multiplication

- a) $425 \times 316 = \mathbf{134\,300}$
b) $583 \times 247 = \mathbf{144\,001}$
c) $714 \times 385 = \mathbf{274\,890}$
d) $836 \times 452 = \mathbf{377\,872}$
e) $947 \times 263 = \mathbf{249\,061}$
f) $652 \times 518 = \mathbf{337\,736}$

2) Replace the letters with numbers

- a) $142\,857 \times 3 = 428\,571$
 $\mathbf{a = 4, b = 2, c = 8, d = 5, e = 7}$
b) $285\,714 \times 3 = 857\,142$
 $\mathbf{f = 8, g = 5, h = 7, i = 1, j = 4}$

The full column-by-column method is on the next page.

Challenge

Omar is **correct**. 999×999 : $8\,991 + 89\,910 + 899\,100 = \mathbf{998\,001}$. Quick check: $999 \times 1000 - 999 = 998\,001$.

Letter-substitution puzzles: how to solve them

Work *right to left*, one column at a time, just like the formal written method.

Each letter stands for a single digit (0–9). The same letter always means the same digit. To solve these puzzles, use the multiplication you already know and work one column at a time, starting from the ones column.

Step 1 — Ones column. Ask: *what digit, multiplied by 3, ends in the ones digit of the answer?* That digit is your first letter. If the product is bigger than 9, carry into the next column.

Step 2 — Move one column to the left. Multiply the next unknown by 3, *add the carry*, and check that the last digit matches the next digit of the answer. That tells you the next letter.

Step 3 — Repeat until you reach the far left. The leftmost column must also work. If it does, you have the answer; if it doesn't, go back and check your carries.

Worked solution for (a):

$$1abcde \times 3 = abcde1$$

Ones column. $3 \times e$ must end in 1. Try: $3 \times 7 = 21$. So $e = 7$, *carry 2*.

Tens column. $3 \times d + 2$ must end in $e = 7$. So $3d$ ends in 5. $3 \times 5 = 15$. $d = 5$, *carry 1*.

Hundreds column. $3 \times c + 1$ must end in $d = 5$. So $3c$ ends in 4. $3 \times 8 = 24$. $c = 8$, *carry 2*.

Thousands column. $3 \times b + 2$ must end in $c = 8$. So $3b$ ends in 6. $3 \times 2 = 6$. $b = 2$, *carry 0*.

Ten-thousands column. $3 \times a$ must end in $b = 2$. $3 \times 4 = 12$. $a = 4$, *carry 1*.

Hundred-thousands column. $3 \times 1 + 1 = 4$, which matches $a = 4$. ✓

$$142857 \times 3 = 428571$$

$$a = 4, b = 2, c = 8, d = 5, e = 7$$

Worked solution for (b):

$$2fghij \times 3 = fghij2$$

Ones column. $3 \times j$ must end in 2. $3 \times 4 = 12$. $j = 4$, *carry 1*.

Tens column. $3 \times i + 1$ must end in $j = 4$. So $3i$ ends in 3. $3 \times 1 = 3$. $i = 1$, *carry 0*.

Hundreds column. $3 \times h$ must end in $i = 1$. $3 \times 7 = 21$. $h = 7$, *carry 2*.

Thousands column. $3 \times g + 2$ must end in $h = 7$. So $3g$ ends in 5. $3 \times 5 = 15$. $g = 5$, *carry 1*.

Ten-thousands column. $3 \times f + 1$ must end in $g = 5$. So $3f$ ends in 4. $3 \times 8 = 24$. $f = 8$, *carry 2*.

Hundred-thousands column. $3 \times 2 + 2 = 8$, which matches $f = 8$. ✓

$$285714 \times 3 = 857142$$

$$f = 8, g = 5, h = 7, i = 1, j = 4$$

A curious fact. The two answers, 142857 and 285714, are made from the same six digits in the same cyclic order. This is not a coincidence: both come from the repeating decimal $\frac{1}{7} = 0.\overline{142857}$. Multiplying 142857 by 2, 3, 4, 5 or 6 always gives a rotation of the same digits.